

A sharp automatic Hessian threshold and complete proximal dynamics for least-squares norm-difference regularization

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Abstract

For least-squares regularization by $\|x\|_1 - \|x\|_p$, we prove that every nonzero local minimum has a positive definite active Hessian when $1 < p \leq 3/2$, assuming only that the sensing matrix has no zero columns. A moment identity excludes null directions below the threshold; at its endpoint the quartic equality case is excluded by a negative sixth-order objective term. For every finite $p > 3/2$, an invertible two-dimensional example has a strict local minimum with singular active Hessian. Strict complementarity and support independence have wider ranges and are stated separately. An explicit common radius and positive step interval certify the complete limiting-subdifferential resolvent, including exclusion of every remote output, equality with the global proximal map, identification, invariance, and variable-step local linear convergence. For a specified $p = 7/4$ example we obtain the sharp true residual exponent $1/3$ and an exact $k^{-1/2}$ asymptotic for the actual complete proximal iteration, with a separate geometric law on the radial axis. No invariance of the radial critical curve is assumed.

1 Model, scope, and relation to earlier work

Let $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$, and $\eta > 0$. Throughout the structural statements, every column of A is nonzero. For $p > 1$ set

$$\Phi_p(x) = \frac{1}{2} \|Ax - b\|_2^2 + \eta(\|x\|_1 - \|x\|_p), \quad F_p = \partial_{\lim} \Phi_p, \quad Q = A^T A, \quad c = A^T b. \quad (1)$$

Here $\partial_{\lim}$ denotes the limiting subdifferential. The least-squares compatibility $c = A^T b$ is retained, including for singular Q . For every $\lambda > 0$ we use the complete relation

$$J_{\lambda F_p}(z) = \{y \in \mathbb{R}^n : z - y \in \lambda F_p(y)\}, \quad r_{F_p}(x) = \text{dist}(0, F_p(x)). \quad (2)$$

Its outputs are unrestricted until an exclusion argument localizes them. The global proximal map is the set of global minimizers of $\Phi_p(y) + \|y - z\|^2 / (2\lambda)$; for a nonconvex function it need not equal (2) without proof. All norms without subscripts on points, matrices, or gradients are Euclidean norms or their operator norms.

The norm-difference model is established prior work. Wang’s penalty formulation includes $\ell_1 - \epsilon \ell_r$ for $r > 1$ [1]. Wang–Zhang study $1 < p \leq 2$ and a smoothed neural optimization method[3]; Huo–Chen–Ge–Ng study $\ell_1 - \beta \ell_q$ and recovery estimates[5]. Support-column independence for the $p = 2$ constrained and penalized models is already given by Yin–Lou–He–Xin, Theorems 2.3–2.4[2]; we do not present support independence as a new standalone principle. Lou–Yan derive a proximal operator and analyze related algorithms for $p = 2$ [4]. The proposed model-specific contribution is the universal automatic active-Hessian threshold and its endpoint mechanism. The quantitative complete-resolvent conclusions are consequences of that structure; ordinary local strong-monotonicity and analytic/KL methods remain applicable where their conditions hold.

Priority boundary. The original published 2015 support-independence proofs have been read. For Wang–Zhang (2017), the publisher abstract, reference list, and Appendix proofs were available, but the main body and theorem statements were not. For Huo et al. (2023), only the publisher abstract/reference list and institutional metadata were available. Accordingly this paper makes no claim of certified first publication of the threshold. The mathematical statements below do not depend on either inaccessible body. The separate source audit records the precise access boundary and comparison obligations; inability to retrieve a source is not evidence that it contains no matching theorem.

2 Automatic structure and the sharp threshold

Theorem 2.1. *For (1) with no zero columns of A , the following statements hold at every nonzero local minimum $\bar{x}$, with $I = \text{supp } \bar{x}$.*

1. *For $1 < p < 2$, strict complementarity is automatic: $|(Q\bar{x} - c)_j| < \eta$ for every $j \notin I$.*
2. *For every $p > 1$, A_I has linearly independent columns.*
3. *For $1 < p \leq 3/2$, the active Hessian*

$$H_I = Q_{II} - \eta \nabla^2 \|\bar{x}_I\|_p \quad (3)$$

is positive definite. In particular $|I| \leq \text{rank } A$.

The exponent $3/2$ in item 3 is sharp across all finite $p > 1$: for each finite $p > 3/2$, there is a two-dimensional example with invertible A and a nonzero strict local minimum whose active Hessian is singular.

Strict complementarity. At a nonzero point the p -norm is C^1 , including at zero coordinates, and the exact smooth sum rule gives

$$F_p(x) = Qx - c + \eta(\partial \|x\|_1 - \nabla \|x\|_p), \quad x \neq 0. \quad (4)$$

Fermat's rule yields $|(Q\bar{x} - c)_j| \leq \eta$ off I . If equality holds, choose $\sigma \in \{-1, 1\}$ so that $\sigma(Q\bar{x} - c)_j = -\eta$, and let $R = \|\bar{x}\|_p > 0$. For $t > 0$ the exact difference is

$$\Phi_p(\bar{x} + \sigma t e_j) - \Phi_p(\bar{x}) = \frac{1}{2} \|A_j\|^2 t^2 - \eta((R^p + t^p)^{1/p} - R).$$

The second term is $-\eta R^{1-p} t^p / p + O(t^{2p})$. For $p < 2$ this is negative for all sufficiently small positive t , contradicting local minimality. $\square$

Support independence. On the fixed active face the objective is analytic and the local-minimum second-order condition gives $H_I \succeq 0$. At a vector all of whose active coordinates are nonzero, the Hessian of the p -norm is positive semidefinite, with kernel precisely the radial line. Indeed, for $R = \|u\|_p$, $w_i = |u_i|^p / R^p$, and $r_i = z_i / u_i$,

$$z^T \nabla^2 \|u\|_p z = (p-1)R \left(\sum_i w_i r_i^2 - \left(\sum_i w_i r_i \right)^2 \right). \quad (5)$$

If $A_I z = 0$ for nonzero z , positivity of H_I and (5) force $z = a\bar{x}_I$ for $a \neq 0$; hence $A\bar{x} = 0$. Dot active stationarity with $\bar{x}$:

$$0 = \|A\bar{x}\|^2 - b^T A\bar{x} + \eta(\|\bar{x}\|_1 - \|\bar{x}\|_p). \quad (6)$$

For $|I| \geq 2$ the last term is positive. For $|I| = 1$, $A\bar{x} = 0$ contradicts the nonzero column assumption. Both cases exclude z . This argument holds for every $p > 1$. $\square$

Positive definiteness up to the endpoint. Suppose that $H_I \succeq 0$ is singular and choose nonzero $z \in \ker H_I$. Apply an orthogonal sign change, assume the active coordinates are positive, and put $x_i = |\bar{x}_i| > 0$, $R = (\sum_i x_i^p)^{1/p}$,

$$w_i = x_i^p / R^p, \quad r_i = z_i / x_i, \quad a = \sum_i w_i r_i, \quad s_i = r_i - a, \quad \mu_k = \sum_i w_i s_i^k.$$

Thus $\mu_1 = 0$. If $\mu_2 = 0$, z is radial; norm homogeneity and $H_I z = 0$ imply $A\bar{x} = 0$, contradicting (6) as above. Hence $\mu_2 > 0$. Factor the radial part exactly, for sufficiently small real t :

$$\frac{\|\bar{x} + tz\|_p}{R} = (1 + at)f\left(\frac{t}{1 + at}\right), \quad f(u) = \left(\sum_i w_i (1 + us_i)^p\right)^{1/p}.$$

Writing $f(u) = 1 + c_2 u^2 + c_3 u^3 + c_4 u^4 + O(u^5)$ gives

$$\begin{aligned} c_2 &= \frac{1}{2}(p-1)\mu_2, & c_3 &= \frac{1}{6}(p-1)(p-2)\mu_3, \\ c_4 &= \frac{1}{24}(p-1)(p-2)(p-3)\mu_4 - \frac{1}{8}(p-1)^3\mu_2^2. \end{aligned} \quad (7)$$

Along the original line the cubic coefficient of the norm is $c_3 - ac_2$. The objective's linear term vanishes by stationarity and its quadratic term vanishes since $z \in \ker H_I$. All coefficients from degree three onwards are minus ηR times the corresponding norm coefficients. Local minimality on both sides therefore requires $a = c_3/c_2$, and the norm's quartic coefficient becomes $C_4 = c_4 - c_3^2/c_2$.

The following exact square identity supplies its sign:

$$\mu_4 - \mu_2^2 - \frac{\mu_3^2}{\mu_2} = \sum_i w_i \left(s_i^2 - \mu_2 - \frac{\mu_3}{\mu_2} s_i \right)^2 \geq 0. \quad (8)$$

Substitution in (7) yields

$$C_4 \geq (p-1) \left(\frac{(3-2p)(p+1)}{24} \mu_2^2 + \frac{(2-p)(p+1)}{72} \frac{\mu_3^2}{\mu_2} \right). \quad (9)$$

For $1 < p < 3/2$ it is strictly positive, so the objective has a negative quartic term, a contradiction.

At $p = 3/2$ the right side is $5\mu_3^2/(576\mu_2)$. If $C_4 > 0$, the same contradiction applies. Otherwise $\mu_3 = 0$, $\mu_4 = \mu_2^2$, and (8) forces $s_i^2 = \mu_2$ at every coordinate. The values $s_i = \pm d$, $d = \sqrt{\mu_2} > 0$, have total weight $1/2$ each, and $a = 0$. The exact norm along the line is consequently

$$\begin{aligned} \frac{\|\bar{x} + tz\|_p}{R} &= \left(\frac{(1+dt)^{3/2} + (1-dt)^{3/2}}{2} \right)^{2/3} \\ &= 1 + \frac{d^2 t^2}{4} + \frac{d^6 t^6}{192} + O(t^8). \end{aligned} \quad (10)$$

Its cubic, quartic, and fifth-order terms vanish. After the quadratic cancellation, the objective has strictly negative sixth-order coefficient $-\eta R d^6/192$, again impossible at a local minimum. Thus $H_I \succ 0$. $\square$

Remark 2.2 (A second proof by radial relaxation). The endpoint sixth-order certificate above is retained. A different fourth-order argument explains its geometric mechanism. For a hypothetical active null direction z , write $\|\bar{x} + tz\|_p = R + n_1 t + n_2 t^2 + n_4 t^4 + O(t^5)$; the cubic vanishes by local minimality and $n_2 > 0$. Put $q_0 = \|A\bar{x}\|^2 > 0$. Homogeneity and $H_I z = 0$ give $\bar{x}^T Q z = 0$, so for $x = (1+s)\bar{x} + tz$,

$$\Phi_p(x) - \Phi_p(\bar{x}) = \frac{q_0 s^2}{2} + \eta n_2 \frac{st^2}{1+s} - \eta n_4 \frac{t^4}{(1+s)^3} + O(t^5).$$

Choosing $s = -\eta n_2 t^2 / q_0$ gives quartic coefficient $-\eta n_4 - \eta^2 n_2^2 / (2q_0)$. Thus a singular active minimum would require $n_4 \leq -\eta n_2^2 / (2q_0) < 0$. The Pearson bound gives $n_4 = RC_4 \geq 0$ for $1 < p \leq 3/2$, a contradiction. The perturbation preserves active signs and leaves inactive coordinates zero.

Proposition 2.3. *For each finite $p > 3/2$, let $R = 2^{1/p}$, $\beta = R(p-1)/2$, $c_4 = -(p-1)(2p-3)(p+1)/24 < 0$, and choose*

$$\alpha > \alpha_0 := \frac{\beta^2}{-4Rc_4} = \frac{3R(p-1)}{2(2p-3)(p+1)}.$$

Let $Q \succ 0$ have eigenvalues α in direction $(1, 1)$ and β in direction $(1, -1)$, and put $A = Q^{1/2}$, $\bar{x} = (1, 1)$, $c = Q\bar{x} + (1 - R/2)(1, 1)$, $b = A^{-T}c$, and $\eta = 1$. Then $\bar{x}$ is a strict local minimum, with singular active Hessian. It is an isolated stationary point. No local true-residual error bound at $\bar{x}$ has exponent greater than $1/3$.

Proof. Stationarity is immediate; the norm Hessian has radial eigenvalue zero and tangent eigenvalue β . Thus the active Hessian eigenvalues are α and zero. Write $x = (1 + h + t, 1 + h - t)$ and let $V(h, t) = \Phi(x) - \Phi(\bar{x})$. On the positive-coordinate neighborhood the norm is analytic for every finite real $p > 1$. Expansion gives

$$V(h, t) = \alpha h^2 + \beta h t^2 - Rc_4 t^4 + O(h^2 t^2 + |h|t^4 + t^6). \quad (11)$$

The analytic implicit function theorem solves $V_h = 0$ as an even curve $h = g(t) = -\beta t^2 / (2\alpha) + O(t^4)$. Its reduced objective is

$$V(g(t), t) = Kt^4 + O(t^6), \quad K = -Rc_4 - \beta^2 / (4\alpha) > 0.$$

For $\alpha = 2\alpha_0$, $K = R(p-1)(2p-3)(p+1)/48 > 0$. Shrink the neighborhood so that $V_{hh} \geq \alpha$ and the reduced objective is at least $Kt^4/2$. Taylor's integral formula gives $V(h, t) \geq \alpha|h-g(t)|^2/2 + Kt^4/2$, proving full-space strictness. Every nearby stationary point lies on g , where the reduced derivative is $4Kt^3 + O(t^5)$; hence only $t = 0$ is stationary after shrinking. Along g , $\|x - \bar{x}\| \sim \sqrt{2}|t|$ while $r_F(x) = |V_t(g(t), t)|/\sqrt{2} = O(|t|^3)$. If $q > 1/3$, the quotient $\|x - \bar{x}\|/r_F(x)^q$ diverges. Stationary isolation makes the same statement valid for distance to the entire stationary set in a sufficiently small ball. This proof gives no uniform exponent for all models with $p > 3/2$. $\square$

Remark 2.4 (Zero columns, the origin, and other stationary points). If $A = [0, 1]$, $b = 0$, and $\eta = 1$, every $(a, 0)$, $a \neq 0$, is a local minimum with zero active Hessian: on its active ray the objective is constant, and off the ray the positive $|y|$ term dominates the negative $O(|y|^p)$ correction. This is a material exception. At the origin, examination of both directions on every coordinate axis shows that local minimality requires $A^T b = 0$. Conversely, if $A^T b = 0$ and A has no zero columns, the origin is the unique global minimum: the norm-difference is nonnegative, and it vanishes only at one-sparse vectors, where the data quadratic is positive. Dotting stationarity with a nonzero x excludes every other stationary point. Formula (4) must not be used at zero. The theorem's quantifier is local minima, not arbitrary stationary points or arbitrary PPA limits. For example at $p = 3/2$ put $R = 2^{2/3}$, $d = R/8$, $\alpha > 0$,

$$Q = \begin{pmatrix} d & -d \\ -d & \alpha + d \end{pmatrix}, \quad c = (1 - R/2, \alpha + 1 - R/2), \quad \bar{x} = (1, 1).$$

Taking $A = Q^{1/2}$ and $b = A^{-T}c$ gives stationarity and $H = \text{diag}(0, \alpha)$, but $\Phi(\bar{x} + te_1) - \Phi(\bar{x}) = Rt^3/32 + O(t^4)$. It is a stationary nonminimum.

Remark 2.5 (The separate strict-complementarity boundary). For every finite $p \geq 2$, take $\eta = 1$, $Q = \text{diag}(1, 2)$, $c = (1, 1)$, $A = \text{diag}(1, \sqrt{2})$, $b = (1, 1/\sqrt{2})$, and $\bar{x} = (1, 0)$. If $r = \|x\|_2$, then

$$\Phi_2(x) - \Phi_2(\bar{x}) = \frac{1}{2}(r - 1)^2 + \frac{1}{2}x_2^2 + (|x_1| - x_1) + (|x_2| - x_2) \geq 0,$$

with equality only at $\bar{x}$. Since $\|x\|_p \leq \|x\|_2$ for $p \geq 2$, the same point is the unique global minimum of Φ_p . Yet $|(Q\bar{x} - c)_2| = \eta$. Thus the automatic strict-complementarity range is $1 < p < 2$, separately from the active-PD threshold $3/2$.

3 An explicit certificate for the complete resolvent

Fix any nonzero local minimum in Theorem 2.1 with $1 < p \leq 3/2$. Let $R = \|\bar{x}\|_p$, $a_* = \min_{i \in I} |\bar{x}_i|$, $s = |I|$, $C_n = n^{1/p-1/2}$, $C_s = s^{1/p-1/2}$, and $h_* = \lambda_{\min}(H_I) > 0$. If $I \neq \{1, \dots, n\}$ set

$$\delta = \min_{j \notin I} \{\eta - |(Q\bar{x} - c)_j|\} > 0, \quad B_Q = \max_{j \notin I} \|Q_j\|.$$

Every condition containing δ or B_Q is omitted for full support. Put

$$r_0 = \min\{a_*/2, R/(2C_n)\}, \quad d = a_*/2, \quad N_0 = R/2, \quad U = \max_{i \in I} |\bar{x}_i| + r_0, \quad W = \sqrt{s} U^{p-1}.$$

For $s \geq 2$, define the finite explicit bound

$$L_H = (p-1)[N_0^{1-p}(2-p)d^{p-3} + (p-1)N_0^{-p}C_s d^{p-2} + 2N_0^{1-2p}W(p-1)d^{p-2} + (2p-1)N_0^{-2p}C_s W^2]. \quad (12)$$

Choose any positive ρ not exceeding each applicable member of

$$r_0, \quad \frac{h_*}{2\eta L_H}, \quad \frac{\delta}{4B_Q}, \quad \frac{R}{2} \left(\frac{\delta}{4\eta} \right)^{1/(p-1)}. \quad (13)$$

The B_Q quotient is omitted if $B_Q = 0$; the L_H quotient is omitted for $s = 1$. Set $m = h_*/2$ for $s \geq 2$, and $m = \|A_I\|^2$ for $s = 1$. Finally set

$$L_p = \sqrt{n} + C_n, \quad M = \|Q\bar{x} - c\| + \|Q\| \rho/4 + \eta L_p. \quad (14)$$

Theorem 3.1. *For any $0 < \lambda_- \leq \lambda_+$ and $r > 0$ satisfying*

$$\lambda_+ M < \rho/2, \quad r < \min\{\rho/4, \lambda_- \delta/2\}, \quad (15)$$

with the second entry omitted for full support, every $\lambda \in [\lambda_-, \lambda_+]$ and every $z \in B_r(\bar{x})$ have exactly one complete resolvent output. It equals the global proximal map, has support I and the signs of $\bar{x}_I$, and obeys

$$\|J_{\lambda F_p}(z) - \bar{x}\| \leq (1 + \lambda m)^{-1} \|z - \bar{x}\| < r. \quad (16)$$

For any $\lambda_k \in [\lambda_-, \lambda_+]$ and $x_0 \in B_r(\bar{x})$, the complete PPA exists uniquely and

$$\|x_k - \bar{x}\| \leq (1 + \lambda_- m)^{-k} \|x_0 - \bar{x}\|.$$

In $B_\rho(\bar{x})$ the strong true-residual error bound is

$$\|x - \bar{x}\| \leq \max\{1/m, 2\rho/\delta\} r_{F_p}(x), \quad (17)$$

with the second constant omitted for full support. In particular $\bar{x}$ is an isolated stationary point and a strict local minimum. All constants are computable for the specified data and minimum; none is uniform over all models or minima.

Continuity and curvature budgets. For $y \in B_\rho(\bar{x})$, $\|y\|_p \geq R - C_n \rho \geq R/2$. Signs of active coordinates are preserved. Off support,

$$|(Qy - c)_j - \eta \nabla_j \|y\|_p| \leq |(Q\bar{x} - c)_j| + B_Q \rho + \eta(2\rho/R)^{p-1} \leq \eta - \delta/2.$$

For active vectors u , with $v_i = \text{sign}(u_i)|u_i|^{p-1}$,

$$\nabla^2 \|u\|_p = (p-1) [\|u\|_p^{1-p} \text{diag}(|u_i|^{p-2}) - \|u\|_p^{1-2p} vv^T].$$

On the active r_0 ball, the diagonal-power difference is bounded by $(2-p)d^{p-3}\|u - \tilde{u}\|$, the vector difference by $(p-1)d^{p-2}\|u - \tilde{u}\|$, and $\|v\| \leq W$. The norm powers have mean-value constants $(p-1)N_0^{-p}C_s$ and $(2p-1)N_0^{-2p}C_s$. Apply these bounds to the two products, with $\|vv^T - \tilde{v}\tilde{v}^T\| \leq 2W\|v - \tilde{v}\|$. This gives exactly (12), hence $H_I(u) \succeq m \text{Id}$ by (13). For $s = 1$ the active norm is linear and its Hessian is identically zero. $\square$

All outputs, independent existence, and dynamics. The penalty $g_p = \|\cdot\|_1 - \|\cdot\|_p$ is globally L_p -Lipschitz. Every Fréchet subgradient has norm at most L_p : combine its lower first-order bound in any unit direction with the upper Lipschitz difference bound. Limiting subgradients inherit this bound. Thus, including at $y = 0$, the smooth sum rule gives $F_p(y) = Qy - c + \eta \partial_{\lim} g_p(y)$ and any complete output satisfies $z = y + \lambda(Qy - c + \eta w)$ with $\|w\| \leq L_p$. Consequently

$$\begin{aligned} (\text{Id} + \lambda Q)(y - z) &= \lambda(c - Qz - \eta w), \\ \|y - z\| &\leq \lambda(\|Qz - c\| + \eta L_p) \leq \lambda M < \rho/2, \end{aligned} \tag{18}$$

because $Q \succeq 0$ and $\|(\text{Id} + \lambda Q)^{-1}\| \leq 1$. Every complete output is therefore in $B_{3\rho/4}(\bar{x})$, where (4) is valid. If $j \notin I$ and $y_j \neq 0$, the offsupport bound and the resolvent equation give

$$\text{sign}(y_j)z_j \geq |y_j| + \lambda\delta/2 > \lambda\delta/2 \geq \lambda_-\delta/2 > r,$$

contradicting $|z_j| \leq \|z - \bar{x}\| < r$. All outputs are identified on the active face. Strong monotonicity on that face, compared to $\bar{x}$, gives (16); comparison of two outputs at the same input gives uniqueness. No existence was presumed in either argument.

Independently, the global proximal objective is continuous and coercive: both data loss and penalty are nonnegative and its added quadratic is coercive. It attains a global minimum. Fermat's rule places every such minimum in the complete resolvent. Thus the resolvent is nonempty and the two maps equal the singleton already proved unique. Invariance and the variable-step estimate follow by induction.

For the residual bound, a nonzero inactive coordinate forces every subgradient's corresponding component to have modulus at least $\delta/2$, giving (17). On the active face all inactive intervals contain zero; the true residual equals the norm of the active gradient, which is at least $m\|x - \bar{x}\|$ by strong monotonicity. This also proves isolation. To prove strictness, take a local-minimum neighborhood. An interior point with the same objective value would itself be a local minimum and hence stationary, contradicting isolation. $\square$

Remark 3.2 (One completely specified positive interval). For the computed ρ, m, δ, M , one may choose $\lambda_+ = \rho/(4M)$, $\lambda_- = \lambda_+/2$, and $r = \min\{\rho/8, \lambda_-\delta/4\}$, omitting the gap entry for full support. These choices satisfy all strict inequalities in (15). No SOSC, residual error bound, or resolvent single-valuedness is assumed in advance.

4 A quartic minimum: sharp true residual and complete PPA

Theorem 4.1. *Fix*

$$p = 7/4, \quad \eta = 1, \quad R = 2^{4/7}, \quad Q = \frac{R}{16} \begin{pmatrix} 19 & 13 \\ 13 & 19 \end{pmatrix}, \quad \bar{x} = (1, 1), \quad c = (1 + 3R/2)(1, 1),$$

and take $A = Q^{1/2}$, $b = A^{-T}c$. In particular $A^T A = Q$, $A^T b = c$, and both columns of A are nonzero. Define on all of $\mathbb{R}^2$

$$\Phi(x) = \frac{1}{2}\|Ax - b\|_2^2 + \|x\|_1 - \|x\|_{7/4}, \quad F = \partial_{\lim} \Phi, \quad S = F^{-1}(0), \quad r_F(x) = \inf_{v \in F(x)} \|v\|_2.$$

The complete resolvent is $J_{\lambda F}(z) = \{y : z - y \in \lambda F(y)\}$, with every output $y \in \mathbb{R}^2$ initially allowed. Put

$$\mathcal{B} = \{x = (1 + h + t, 1 + h - t) : |h|, |t| \leq 1/32\}, \quad \mathcal{U} = \overline{B}_{1/64}(\bar{x}).$$

The following assertions hold.

1. $\bar{x}$ is a strict local minimum and is the sole stationary point in $\mathcal{B}$. The active Hessian at $\bar{x}$ has eigenvalues $2R$ and 0 .
2. On $\mathcal{B}$ the **strong true-residual error bound** is

$$\|x - \bar{x}\|_2 \leq (512/R)^{1/3} r_F(x)^{1/3}.$$

The exponent $1/3$ is optimal: no exponent greater than $1/3$, with any finite constant and any neighborhood of $\bar{x}$, satisfies this bound. On $\mathcal{U}$, $d(x, S) = \|x - \bar{x}\|_2$.

3. For **every** $0 < \lambda \leq 1/128$ and **every** $z \in \mathcal{U}$, the complete $J_{\lambda F}(z)$ is a singleton, equals the global minimizing proximal map

$$\operatorname{argmin}_{y \in \mathbb{R}^2} \left\{ \Phi(y) + \frac{\|y - z\|_2^2}{2\lambda} \right\},$$

and belongs to $\mathcal{U}$.

4. Fix any one $\lambda \in (0, 1/128]$, and run the ordinary complete PPA from **any** $x_0 \in \mathcal{U}$: $x_{k+1} \in J_{\lambda F}(x_k)$. This trajectory is unique and converges to $\bar{x}$. Writing $x_k = (1 + h_k + t_k, 1 + h_k - t_k)$, if $t_0 \neq 0$, then t_k keeps its initial sign, and

$$\frac{h_k}{t_k^2} \longrightarrow -\frac{3}{32}, \quad \sqrt{k}|t_k| \longrightarrow \sqrt{\frac{128}{13\lambda R}}, \quad \boxed{\sqrt{k}\|x_k - \bar{x}\|_2 \longrightarrow \frac{16}{\sqrt{13\lambda R}}}.$$

If $t_0 = 0$, the exact separate formula is $t_k = 0$, $h_k = (1 + 2\lambda R)^{-k} h_0$.

The constants $1/64$ and $1/128$ below are certified sufficient values, not maximal basin or step bounds. The asymptotic assertions fix λ ; they do not assert this particular constant for an arbitrary variable-step sequence.

4.1 Exact scalar formula and analytic remainder certificate

Throughout $\mathcal{B}$, the two coordinates of x are positive. Put $s = 1 + h$, $u = t/s$, and

$$f(u) = \left(\frac{(1+u)^{7/4} + (1-u)^{7/4}}{2} \right)^{4/7}, \quad \varphi(h, t) = \Phi(1+h+t, 1+h-t) - \Phi(\bar{x}).$$

The eigenvalues of Q are $2R$ in the radial direction $(1, 1)$ and $3R/8$ in the tangent direction $(1, -1)$. Substitution, including $c = A^T b$, gives the exact identity

$$\frac{\varphi(h, t)}{R} = 2h^2 + \frac{3}{8}t^2 - s(f(u) - 1). \quad (\text{D.1})$$

In particular $\nabla \Phi(\bar{x}) = 0$. The symmetric binomial expansion gives

$$f(u) = 1 + \frac{3}{8}u^2 - \frac{11}{256}u^4 + E(u). \quad (\text{D.2})$$

Here, for **all real** $|u| \leq 1/16$,

$$|E(u)| \leq 5|u|^6, \quad |E'(u)| \leq 28|u|^5, \quad |E''(u)| \leq 140|u|^4. \quad (\text{D.3})$$

For completeness these bounds have an analytic certificate, rather than a sampled verification. Let $G(u) = ((1+u)^{7/4} + (1-u)^{7/4})/2$. Its even binomial coefficients from degree two onward are positive; after degree four their absolute values decrease. For complex $|u| \leq r = 7/8$,

$$|G(u) - 1| \leq \frac{21}{32}r^2 + \frac{35}{2048} \frac{r^4}{1-r^2} = \frac{214375}{393216} < 1.$$

Therefore the analytic branch $G^{4/7}$ with value one at zero is defined on a neighborhood of this closed disk and has modulus at most 2. If $f = \sum a_{2j}u^{2j}$, the Cauchy coefficient estimate is $|a_{2j}| \leq 2r^{-2j}$. After removal of the displayed terms through degree four, put $v = (u/r)^2 \leq 1/196$. Summing the majorant series and its two derivatives gives

$$\begin{aligned} |E|/|u|^6 &\leq 2r^{-6}/(1-v) < 5, \\ |E'|/|u|^5 &\leq 2r^{-6} [6/(1-v) + 2v/(1-v)^2] < 28, \\ |E''|/|u|^4 &\leq 2r^{-6} [30/(1-v) + 26v/(1-v)^2 + 8v^2/(1-v)^3] < 140. \end{aligned}$$

The inequalities at zero are interpreted by continuity. This proves (D.3) on the full interval. Direct expansion of (D.1) consequently gives

$$\frac{\varphi}{R} = 2h^2 + \frac{3}{8}ht^2 + \frac{11}{256}t^4 + O(h^2t^2 + |h|t^4 + t^6). \quad (\text{D.4})$$

4.2 Star inequality, strictness, and the sharp residual exponent

The quantity needed for invariance is the gradient inner product with the displacement, not the objective in (D.4). Exactly,

$$D(h, t) := \langle x - \bar{x}, \nabla \Phi(x) \rangle = h\varphi_h + t\varphi_t.$$

Differentiating (D.1) and using (D.2) gives

$$\frac{D}{R} = 4h^2 + \frac{9}{8}hu^2 + \frac{3}{4}h^2u^2 + \left(\frac{11}{64} + \frac{11h}{256} \right) u^4 - hE(u) - uE'(u). \quad (\text{D.5})$$

On $\mathcal{B}$, $|u| \leq 1/31 < 1/16$, so (D.3) applies. Complete the first square in (D.5), discard the nonnegative term $3h^2u^2/4$, and bound the remaining terms:

$$\frac{D}{R} \geq 4 \left(h + \frac{9}{64} u^2 \right)^2 + \left[\frac{95}{1024} - \frac{11}{8192} - \frac{28 + 5/32}{961} \right] u^4 \geq 4 \left(h + \frac{9}{64} u^2 \right)^2 + \frac{1}{32} u^4. \quad (\text{D.6})$$

The last rational bracket exceeds $1/32$. Also $(33/32)^4 < 2$, so

$$h^2 + t^4 \leq 2 \left(h + \frac{9}{64} u^2 \right)^2 + \left(2 \left(\frac{9}{64} \right)^2 + 2 \right) u^4.$$

Comparison with (D.6) yields the conservative full-box bound

$$\boxed{D(h, t) \geq \frac{R}{128} (h^2 + t^4).} \quad (\text{D.7})$$

One can sharpen (D.7) to $R(h^2 + t^4)/64$ by keeping $(33/32)^4$ rather than replacing it by two: the required rational check is $(2(9/64)^2 + (33/32)^4)/64 < 1/32$. We use this certified sharper bound below:

$$D(h, t) \geq \frac{R}{64} (h^2 + t^4). \quad (\text{D.8})$$

Every ray segment from $\bar{x}$ to a point of $\mathcal{B}$ stays in $\mathcal{B}$. Integrating (D.8) along that segment gives

$$\varphi(h, t) \geq \frac{R}{128} h^2 + \frac{R}{256} t^4 > 0 \quad \text{when } (h, t) \neq (0, 0).$$

This proves strict local minimality. A stationary point in the box has $D = 0$, so (D.8) proves stationarity isolation there. The Hessian eigenvalues stated in the theorem follow from (D.1) or from cancellation of the tangent norm Hessian against $3R/8$.

Since $\mathcal{B}$ has positive coordinates, $F(x) = \{\nabla \Phi(x)\}$ there: this is the actual complete residual. Put $d = \|x - \bar{x}\|_2 = \sqrt{2(h^2 + t^2)}$. Since $|h| < 1$,

$$h^2 + t^4 \geq h^4 + t^4 \geq \frac{1}{2} (h^2 + t^2)^2 = d^4/8.$$

Consequently $dr_F(x) \geq D(h, t) \geq Rd^4/512$, proving the strong exponent-1/3 error bound, including $d = 0$. Every other stationary point is outside $\mathcal{B}$, hence at distance greater than $\sqrt{2}/32$ from $\bar{x}$. For $x \in \mathcal{U}$, its distance to such a point exceeds $\sqrt{2}/32 - 1/64 > 1/64 \geq d$, proving the asserted equality with distance to the entire S .

To prove optimality, $\varphi_{hh}(0, 0) = 4R > 0$, and the analytic implicit function theorem gives the unique even radial critical curve near zero:

$$\varphi_h(h_*(t), t) = 0, \quad h_*(t) = -\frac{3}{32} t^2 + O(t^4).$$

Substitution into (D.4) gives

$$\varphi(h_*(t), t) = \frac{13R}{512} t^4 + O(t^6), \quad \varphi_t(h_*(t), t) = \frac{13R}{128} t^3 + O(t^5).$$

The exact gradient-coordinate transformation is $\|\nabla \Phi\|_2 = \sqrt{\varphi_h^2 + \varphi_t^2}/\sqrt{2}$. Thus on this curve $d \sim \sqrt{2}|t|$ whereas $r_F \sim 13R|t|^3/(128\sqrt{2})$. For any $q > 1/3$, d/r_F^q tends to infinity. The critical curve is used only to test sharpness; it is not declared invariant under PPA.

4.3 Complete-fiber exclusion, global proximal equality, and invariance

Let $g(x) = \|x\|_1 - \|x\|_{7/4}$. It is globally Euclidean-Lipschitz with constant $L_p = \sqrt{2} + 2^{1/14}$. Every Fréchet subgradient has norm at most L_p , by testing the subgradient inequality in unit directions against the Lipschitz upper bound; every limiting subgradient inherits this bound. The exact smooth-sum rule, valid including at zero, is

$$F(y) = Qy - c + \partial_{\lim} g(y).$$

This does not assert the invalid norm-gradient formula at zero. Given **any** complete resolvent output y at z , choose $w \in \partial_{\lim} g(y)$ with $z - y = \lambda(Qy - c + w)$. Then

$$(I + \lambda Q)(y - z) = \lambda(c - Qz - w), \quad \|y - z\|_2 \leq \lambda(\|Qz - c\|_2 + L_p), \quad (\text{D.9})$$

because $Q \succ 0$. The elementary bounds

$$7/5 < R < 3/2, \quad \sqrt{2} < 10/7, \quad 2^{1/14} < 15/14$$

give, for $z \in \mathcal{U}$,

$$\|Qz - c\|_2 + L_p \leq \sqrt{2}(1 - R/2) + 2R/64 + L_p < \frac{3}{7} + \frac{3}{64} + \frac{5}{2} < 3. \quad (\text{D.10})$$

Therefore **every** complete output satisfies

$$\|y - \bar{x}\|_2 < 1/64 + 3/128 = 5/128, \quad |h_y|, |t_y| < 5/(128\sqrt{2}) < 1/32. \quad (\text{D.11})$$

No localization of outputs was assumed in deriving (D.9). In particular remote nonsmooth outputs and zero are excluded before using any smooth formula.

On the convex box $\mathcal{B}$, both individual coordinates are at least $15/16$. The usual norm Hessian formula gives

$$0 \preceq \nabla^2 \|y\|_{7/4} \preceq \frac{3}{4} \|y\|_{7/4}^{-3/4} \text{diag}(y_i^{-1/4}).$$

Here $\|y\|_{7/4} \geq (15/16)R > 1$ and $(3/4)(16/15)^{1/4} < 1$. Thus $\nabla^2 \Phi(y) = Q - \nabla^2 \|y\|_{7/4} \succ -I$. The map $y \mapsto y + \lambda \nabla \Phi(y)$ is strictly monotone on this convex box, because its Jacobian is at least $(1 - \lambda)I \succ 0$. All complete outputs from (D.11) lie in this box, so two such outputs at the same input must coincide.

For existence, Φ is continuous and nonnegative; hence its proximal objective is continuous and coercive for every input and every positive step. A global minimizer exists, and Fermat places every global minimizer in the complete $J_{\lambda F}$. The proved at-most-one output therefore gives exactly one complete output and exactly one global minimizer, and the two maps coincide.

Finally, write $z = y + \lambda \nabla \Phi(y)$. By (D.8),

$$\|z - \bar{x}\|_2^2 = \|y - \bar{x}\|_2^2 + 2\lambda D(h_y, t_y) + \lambda^2 \|\nabla \Phi(y)\|_2^2 \geq \|y - \bar{x}\|_2^2. \quad (\text{D.12})$$

Thus $y \in \mathcal{U}$, establishing the stated invariant ball for the complete resolvent and global proximal map.

4.4 Actual PPA tracking and the exact slow-tail constant

Fix $0 < \lambda \leq 1/128$. By complete coverage, uniqueness, and (D.12), the trajectory starting at any $x_0 \in \mathcal{U}$ is defined uniquely for every integer $k \geq 0$ and stays in that ball. Summing (D.12) and using (D.8) gives

$$\sum_{k=0}^{\infty} (h_{k+1}^2 + t_{k+1}^4) < \infty.$$

Therefore $h_k, t_k \rightarrow 0$, and $x_k \rightarrow \bar{x}$. This argument uses the ordinary proximal equations, not a reduced scalar minimization algorithm.

The exact coordinate form of those equations is

$$h_k = h_{k+1} + \frac{\lambda}{2} \varphi_h(h_{k+1}, t_{k+1}), \quad t_k = t_{k+1} + \frac{\lambda}{2} \varphi_t(h_{k+1}, t_{k+1}). \quad (\text{D.13})$$

The factor $1/2$ comes from the Euclidean metric $\|x - x_k\|_2^2 = 2[(h - h_k)^2 + (t - t_k)^2]$. From (D.1)–(D.3), on the box,

$$\begin{aligned} \varphi_h/R &= 4h + \frac{3}{8}t^2 + O(|h|t^2 + t^4), \\ \varphi_t/R &= \frac{3}{4}ht + \frac{11}{64}t^3 + O(h^2|t| + |h||t|^3 + |t|^5). \end{aligned} \quad (\text{D.14})$$

All remainders here are analytic with uniform bounds on the box; these are local asymptotic estimates derived from the explicit certificate, not assumptions about the trajectory.

If $t \neq 0$, its exact tangent derivative is

$$\frac{\varphi_t}{Rt} = \frac{3}{4} \frac{h}{s} + \frac{11}{64} \frac{t^2}{s^3} - \frac{E'(t/s)}{t}.$$

By (D.3), $|h|, |t| \leq 1/32$, and $s \geq 31/32$, the absolute value is at most

$$\frac{3}{124} + \frac{11}{64 \cdot 32^2} (32/31)^3 + \frac{28}{32^4} (32/31)^5 < \frac{1}{32}. \quad (\text{D.15})$$

The bound extends continuously to $t = 0$. Since $\lambda R/64 < 1$, the factor in the second equation of (D.13), $1 + (\lambda/2)(\varphi_t/t)$, is positive everywhere in the box. Consequently $t_{k+1} = 0$ if and only if $t_k = 0$, and nonzero tangent coordinates keep their sign. On the axis $t = 0$, (D.1) gives $\varphi = 2Rh^2$, so (D.13) gives exactly the geometric formula in the theorem.

Now suppose $t_0 \neq 0$. The tangent multiplier tends to one as $h_k, t_k \rightarrow 0$, and therefore

$$\frac{t_k}{t_{k+1}} \rightarrow 1. \quad (\text{D.16})$$

Set $a = 1 + 2\lambda R > 1$, $b_0 = 3\lambda R/16$, and $q = 1/a < 1$. Introduce the analytic even function

$$\kappa(u) = \frac{1 - f(u) + uf'(u)}{u^2}, \quad \kappa(0) = 3/8.$$

The value at zero is its removable analytic extension. Exact differentiation of (D.1) gives $\varphi_h/R = 4h + u^2\kappa(u)$. For the actual output with $s_{k+1} = 1 + h_{k+1}$, the first equation of (D.13) therefore yields the exact tracking recurrence

$$\begin{aligned} w_{k+1} &= q \left(\frac{t_k}{t_{k+1}} \right)^2 w_k - q \frac{\lambda R}{2} \frac{\kappa(t_{k+1}/s_{k+1})}{s_{k+1}^2}, \\ w_k &= h_k/t_k^2. \end{aligned} \quad (\text{D.17})$$

No boundedness of w_k is presumed. By (D.16) and convergence to the origin, the multiplier tends to $q < 1$ and the forcing tends to $-qb_0$. Choose any $\theta \in (q, 1)$; eventually the multiplier has absolute value below θ and the forcing is bounded. Iterating this scalar inequality first proves w_k bounded. Subtract the fixed point $-qb_0/(1 - q)$: the new forcing tends to zero, and the same contraction then proves convergence. Hence

$$\frac{h_k}{t_k^2} \rightarrow -\frac{qb_0}{1 - q} = -\frac{3}{32}. \quad (\text{D.18})$$

This is the required radial-following proof. It is valid for all off-axis initial points in the certified ball; it neither assumes nor requires invariance of the radial critical curve.

Substitute (D.18) into the second equation of (D.13), using (D.14) at the actual output. The remainders become $o(t_{k+1}^2)$ after division by t_{k+1} , and

$$t_k = t_{k+1} \left[1 + \frac{13\lambda R}{256} t_{k+1}^2 + o(t_{k+1}^2) \right]. \quad (\text{D.19})$$

Writing $\tau_k = |t_k| > 0$, expansion of the reciprocal square gives

$$\tau_{k+1}^{-2} - \tau_k^{-2} \longrightarrow \frac{13\lambda R}{128}.$$

Telescoping and taking Cesàro means yields $\tau_k^{-2}/k \rightarrow 13\lambda R/128$, proving the tangent limit. Finally (D.18) implies $h_k/t_k \rightarrow 0$, and $\|x_k - \bar{x}\|_2 = \sqrt{2}\sqrt{h_k^2 + t_k^2} \sim \sqrt{2}\tau_k$. This proves the stated exact distance limit.

5 Interpretation and verification boundaries

The structural transition concerns active curvature at every local minimum. It does not imply that all stationary points are minima, that arbitrary PPA initializations reach a minimum, or that a sparse ground-truth signal is recovered. Nondegenerate local rates have their ordinary strong-monotonicity interpretation. The quartic example also admits the classical analytic gradient/KL interpretation. Here its complete-map certification and exact actual-trajectory asymptotic distinguish the algorithmic assertion from a reduced scalar curve calculation. No exponent or tail is asserted uniformly across all upper-side models. The step and basin constants are sufficient rather than optimal.

The accompanying Python checks verify exact finite rational identities, analytic-majorant arithmetic, high-precision boundary expansions, and actual two-variable implicit proximal equations. These corroborate the analytic proofs, which carry the universal claims. The central publication-priority gate remains subject to the two explicit primary-text access gaps in Section 1. An exact antecedent, if found, must be credited. Renaming standard local consequences as PPA cannot replace lost novelty.

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